

Multimodal Generative AI Models: Towards Perceptually-grounded NLP Models

图文讲义版：用原 PPT 作为视觉锚点，按“概念故事线 + 直觉解释 + 考试速记”整理。

根本问题：只在符号里解释符号会陷入 Chinese Room / dictionary-go-round，模型可能会操作词形但没有 referent → grounding 目标：perceptually grounded NLP 要让符号能通过感知、分类和交互指向世界中的对象、属性与事件 → world scopes：学习者能访问的数据范围决定能学到什么，图像、行动和社会互动逐层改变 reporting bias → VLM 定义：Vision+Language Models 把视觉和语言数据结合起来，支持 captioning、VQA、visual grounding 等任务 → 视觉表示：从 CNN 的 pooled/spatial features 到 Vision Transformer 的 image patches as tokens，视觉也要变成模型可处理的表示 → 早期任务模型：image captioning / VQA 通常用视觉 encoder 加语言 decoder 或分类头，但视觉信息可能只在开头注入 → 通用架构选择：Transformer-based VLM 主要通过 cross-encoding、cross-attention 或 visual prefix 融合语言 token 与视觉 token → 训练目标：VLM 需要 LM、MIM、ITM、MMM 等 loss，让模型同时学视觉、语言、匹配和跨模态补全 → 评测转向：不能只看下游任务分数，还要用 FOIL、SVO-Probes、Winoground、intrinsic evaluation 检查是否真的理解组合关系与表示质量

Symbol Grounding Problem

How can the meanings of the **meaningless symbol tokens**, manipulated solely on the basis of their (arbitrary) **shapes**, be grounded in anything but other **meaningless symbols**?

Harnad, Stevan. "The symbol grounding problem." Physica D: Nonlinear Phenomena 42.1-3 (1990): 335-346.

这节课一句话

这节课一句话：text-only NLP 像只读菜单却从没见过菜，Multimodal / Vision+Language Models 的核心任务就是把语言符号重新接到视觉、经验和世界上，同时还要设计能融合、训练、评测这些跨模态表示的模型机制。

1. Symbol grounding: 为什么只读文字不等于理解世界

Symbol Grounding Problem

How can the meanings of the **meaningless symbol tokens**, manipulated solely on the basis of their (arbitrary) **shapes**, be grounded in anything but other **meaningless symbols**?

Hamad, Stevan. "The symbol grounding problem." Physica D: Nonlinear Phenomena 42.1-3 (1990): 335-346.

人话直觉

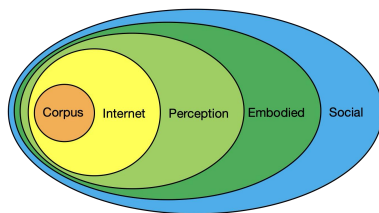
先把 text-only model 想成一个被关在图书馆里的学生：它可以背下“pasticciotto 是一种甜点”“甜点是饭后吃的食物”，但如果从没看过、摸过、吃过，它的意义就像一串互相解释的便利贴。便利贴越贴越多，看起来很博学，但最底下没有桌子撑住。Symbol grounding problem 问的正是：这些符号到底怎样碰到世界？

精确定义 / 机制： Symbol grounding problem 指的是：如果系统只根据符号的任意形状和符号之间的关系来操作 tokens，那么这些本来无意义的 tokens 的意义如何不只是由另一批无意义 tokens 定义？Chinese Room argument 强调，按规则正确操作符号不必然等于理解；Chinese/Chinese dictionary-ground 则说明，如果学习中文只能查中文字典，解释会在中文词之间循环，无法接到外部经验。Perceptually grounded NLP 的目标不是否定语言中的分布式规律，而是补上 referent：系统要能通过 perception / categorisation / sensorimotor interaction 把符号接到对象、属性、动作、空间关系和事件上。

考试问法：如果考“为什么需要 multimodal models / what is symbol grounding problem”，不要只说 text-only 模型数据少。高分答案要写：纯文本学习容易让意义停留在 symbol-symbol relation；grounding 要把 symbols 连接到 perception、categorisation 和 real-world referents；视觉模态提供物体、属性、空间与动作信号，因此能缓解但不自动解决 grounding。

2. World scopes: 模型住在哪个世界，就只能学到那个世界的常识

Experience Grounds Language



Each layer is a **world scope** =
Describes the “data” that the
learner has access to

How does the **reporting bias** change?
What **new knowledge** do models have access to?
What advances in **modelling** and **fusion** are necessary?

Bisk, Y., Holtzman, A., Thomason, J., Andreas, J., Bengio, Y., Chai, J., ... & Turian, J. (2020, November). Experience Grounds Language. EMNLP 2020

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人话直觉

World scope 像给学习者开地图权限。只给文字，它像听收音机认识世界；给图像，它终于能看到物体长什么样；允许行动，它开始知道推杯子会倒、开门会进房间；允许合作，它才学到语言里的请求、承诺和共同目标。权限每升一级，数据分布就变了，模型能学到的常识也变了。

精确定义 / 机制： Experience Grounds Language 的核心观点是：语言学习依赖 learner access to experience。每一层 world scope 描述学习者可访问的数据和环境范围，也改变 reporting bias。Reporting bias 是文本世界的偏差：人们通常不写“香蕉是黄色的”“杯子会挡住桌面”这类显而易见的感知事实，却会写异常、有趣或值得报道的内容。因此，text-only data 不是世界本身，而是人类选择性说出来的世界。引入图像、视频、行动和互动，会让模型接触到文本不常说但感知中稳定存在的信息，也迫使模型需要新的 modelling and fusion mechanisms。

考试问法：如果考 world scope，按三步答：定义它是 learner 可访问的数据/环境范围；解释更高 scope 会带来新知识并改变 reporting bias；说明 VLM 至少把模型从纯文本 scope 推向 vision+language scope，但离完整 embodied / social grounding 仍有距离。

3. VLM 任务地图：说图、问图、找图中被语言指到的东西

Vision+Language Models Tasks



- Image captioning
 - A group of people eating noodles
- Visual Question Answering
 - Q: What are the people eating?
 - A: Noodles
- Visual grounding
 - Given a language description, identifies the associated target object

人话直觉

Vision+Language task 可以想成三种课堂小测：老师给你一张图，让你描述，这是 image captioning；老师指着图问“他们在吃什么？”，这是 VQA；老师说“戴红帽子的男孩”，让你在图中圈出来，这是 visual grounding。三题合起来测试的不是背单词，而是语言能不能落到图像中的对象和关系。

精确定义 / 机制： Vision+Language Model 整合视觉和语言数据，以完成理解、生成和决策任务。Lecture 25 的基础任务包括 image captioning，即给定图像生成自然语言描述；Visual Question Answering，即输入图像和问题并输出答案；visual grounding / referring expression understanding，即根据语言描述识别图像中的目标对象。后续数据集还包括 MSCOCO captioning、VQA dataset、referring expressions、Visual Storytelling 和 GuessWhat?! visual dialogue。它们的共同目标是测试模型是否能把 textual expressions 对齐到 visual content，而不仅是利用语言 prior 猜答案。

考试问法： 如果考“list and define VLM tasks”，最稳结构是逐个定义：captioning = image to text；VQA = image + question to answer；visual grounding/referring expression = language description to target object/bounding box；再补 visual storytelling 和 visual dialogue 作为扩展任务。

4. 视觉表示：从 CNN 特征图到 Vision Transformer 的图像 tokens

Vision Transformers



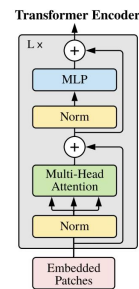
Dosovitskiy, Alexey. "An image is worth 16x16 words: Transformers for image recognition at scale." arXiv preprint arXiv:2010.11929 (2020).
Image credit: <https://www.7tubs.com/blog/vision-transformer-guide>

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人话直觉

语言模型吃 tokens，但图像原本是一大片像素。VLM 的第一步像给图像切寿司：CNN 会先从像素里卷出边缘、纹理、部件等 feature maps；Vision Transformer 更直接，把图像切成小方块 patches，每块压成一个 token，于是图像也能排成一串“视觉句子”。

精确定义 / 机制：早期 VLM 常用 CNN 表示视觉信息：卷积层从数值像素中学习局部、层级特征，得到 spatial feature maps 或 pooled visual features。Vision Transformer 的关键转变是把 image split into patches，每个 patch 线性投影成 embedding，加上位置编码后交给 Transformer 处理。这个设计对 multimodal models 很重要，因为主流语言模型和 Transformer VLM 都擅长处理离散 token 序列；当图像变成 visual tokens 后，语言 token 与视觉 token 才能通过 attention、cross-attention 或 prefix 机制在同一计算框架里交互。

考试问法：如果考“how represent visual information in VLMs”，要比较 CNN 和 ViT：CNN 学 feature maps / pooled features；ViT 把图像切成 patches as tokens；二者目的都是把视觉输入转换成可与语言模型融合的 dense representations。

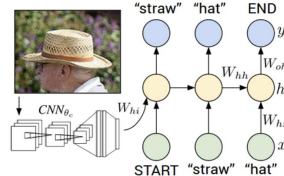
5. 早期 captioning: 只在开头看一眼图, 语言生成会越跑越远

Task-specific models: Image captioning

A language model conditioned on visual information implemented as an **encoder-decoder** neural architecture

Model: uses a CNN to extract a pooled representation that it is used to **initialise** the hidden state of a language model RNN

- The RNN generates tokens one by one conditioned on the previous tokens
- **Note:** conditioning on the image happens only at the **very beginning** of the sequence
- This could be **problematic**; why?



Karpathy, A., & Fei-Fei, L. (2015). Deep visual-semantic alignments for generating image descriptions. In Proceedings of the IEEE conference on computer vision and pattern recognition (pp. 3128-3137).

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人话直觉

早期 image captioning 像让学生看 3 秒图片, 然后闭眼写作文。开头可能还记得“有一群人在吃面”, 但写着写着, 语言模型自己的惯性会接管: 它更会续写常见句子, 而不是持续检查图里到底有什么。

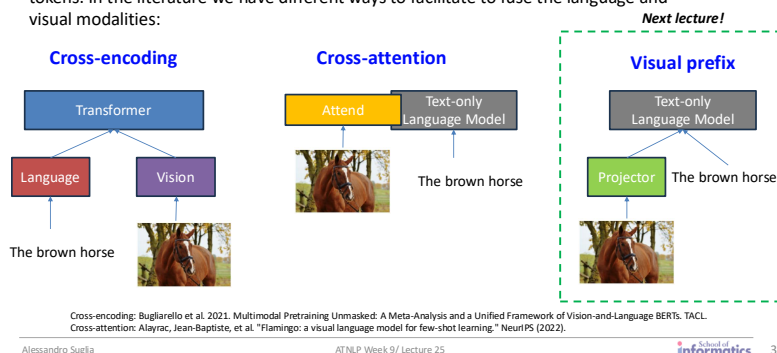
精确定义 / 机制: Task-specific image captioning 模型通常是 encoder-decoder neural architecture: CNN 作为 visual encoder 提取 pooled image representation, 再用这个表示初始化 RNN language model 的 hidden state; RNN conditioned on previous tokens 一个词一个词生成 caption。问题在于 conditioning on image happens only at the very beginning of the sequence。随着生成步数增加, 视觉信号可能被 hidden state 稀释, 模型更依赖语言 prior, 导致 hallucinated objects、遗漏细节或图文对齐变弱。这也是后来 attention-based captioning、cross-modal fusion 和 Transformer VLM 要解决的核心痛点。

考试问法: 如果考“early image captioning architecture and limitation”, 按 encoder-decoder 讲: CNN extracts pooled visual representation → initialise RNN hidden state → RNN generates tokens autoregressively。局限是 image conditioning 只在开头, 后续视觉 grounding 可能衰减。

6. 三种融合路线：一起编码、文本看图、把图变成前缀

Main Approaches for Designing Transformer-based VLMs

Most popular VLM architectures rely on Transformers which operate on discrete tokens. In the literature we have different ways to facilitate to fuse the language and visual modalities:



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人话直觉

把视觉和语言融合有三种像样的聚会方式。Cross-encoding 是把图片和文字都请进同一个房间，从头一起聊天；cross-attention 是语言模型坐在主桌，需要时转头问视觉专家；visual prefix 则更像把图片翻译成一串特殊提示词，塞到 LLM 句子开头，让它以为自己读到了一段视觉上下文。

精确定义 / 机制：Transformer-based VLM 的核心设计问题是 how to fuse language and visual modalities。Cross-encoding 把 language tokens 和 vision tokens 一起输入同一个 Transformer，让 self-attention 在统一序列中学习跨模态关系，适合理解和匹配。Cross-attention 保留 text-only language model 的主体，让语言 queries attend to visual representations，典型思路是模型生成语言时可以动态读取视觉信息。Visual prefix 把视觉 encoder 输出通过 projector 映射成 LLM 可接受的 prefix tokens，再让 frozen 或 partly tuned LLM 在这些视觉 tokens 条件下生成。三者共同目标都是让语言处理过程能访问视觉信息，但计算成本、可扩展性、对预训练 LLM 的复用程度不同。

考试问法：如果考“compare VLM architecture choices”，一定写共同目标：fuse visual and linguistic information。再区分机制：cross-encoding = joint Transformer over both tokens；cross-attention = language model attends to visual features；visual prefix = projected visual embeddings prepended as prefix tokens。

7. 训练目标：VLM 不是只学说话，还要学看图和配对

Training losses for V+L models

Optimise
loss

- L_{MIM} : **Masked Image Modelling** = Randomly mask some patches in the image, and ask to reconstruct them (e.g., predict a tensor of the same size of the vision tokens and use MSE loss)
- L_{LM} : **Language Modelling** = Classic language modelling loss or masked language modelling (like BERT)
- L_{ITM} : **Image-Text Modelling** = Given an image and a caption, decide whether the image matches the caption (i.e., binary classification)
- L_{MMM} : **Masked Multimodal Modeling** = Simultaneously mask both vision and language, and predict both of them

Singh, A., Hu, R., Goswami, V., Couairon, G., Galuba, W., Rohrbach, M., & Kiela, D. (2022). Flava: A foundational language and vision alignment model. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (pp. 15638-15650).

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人话直觉

训练 VLM 像训练一个双语导游：只练说话，它可能口才好但看不懂展品；只练看图，它不会解释给游客听；只练图文配对，它会判断“像不像”，但不一定会生成。不同 loss 就是在练不同肌肉。

精确定义 / 机制：V+L models 通常组合多种 training losses。Masked Image Modelling 随机遮蔽 image patches，让模型重构视觉 token 或特征，训练视觉补全能力；Language Modelling 使用 next-token prediction 或 masked LM，训练语言理解/生成；Image-Text Matching 给定 image-caption pair 判断是否匹配，是二分类式跨模态对齐；Masked Multimodal Modelling 同时 mask vision and language，让模型利用两个模态互相补全。多 loss 的必要性在于 VLM 需要同时掌握单模态表示、跨模态 alignment 和 fused reasoning。

$$\mathcal{L}_{VLM} = \lambda_{MIM}\mathcal{L}_{MIM} + \lambda_{LM}\mathcal{L}_{LM} + \lambda_{ITM}\mathcal{L}_{ITM} + \lambda_{MMM}\mathcal{L}_{MMM}$$

考试问法：如果考 training objectives，列名字还不够，要说明每个 loss 训练什么：MIM 训练视觉重构，LM 训练语言，ITM 训练图文匹配，MMM 训练跨模态补全。可补充 ITM 与 contrastive learning 的区别：ITM 常是 match/non-match binary classification，contrastive learning 是在 batch 内拉近正对、推远负对。

8. 评测陷阱：会答数据集题，不代表真的懂图文关系

Winoground

- 1,600 text-image pairs to evaluate compositional understanding
- Examples curated by human experts to assess **visual grounding** skills
 - Curated in such a way that captions have the same words but appear in different order
- Task: match the correct caption to its image
- Models struggle, often **performing below chance**



(a) there is [a mug] in [some grass]



(c) a person [sits] and a dog [stands]



(b) there is [some grass] in [a mug]



(d) a person [stands] and a dog [sits]

Thrush et al. (2022). Winoground: Probing vision and language models for visio-linguistic compositionality. CVPR.

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人话直觉

很多 VLM 像会刷题的学生：在常见 VQA 或 captioning benchmark 上分数不错，但一遇到“狗追人”和“人追狗”这种词都一样、关系反了的题，就露馅。Winoground 这类测试就是把捷径堵上，逼模型理解组合关系，而不是数图里有什么名词。

精确定义 / 机制： VLM evaluation 分为 extrinsic 和 intrinsic 两大方向。Extrinsic evaluation 用具体任务/数据集评估应用表现，例如 captioning、VQA 或 multi-task benchmark；优点是指标明确，缺点是可能只反映某个数据集上的 shortcut。Challenge datasets 更专门测试特定能力：FOIL captions 用 contextually plausible but incorrect image-text pairs 测 noun-level mismatch；SVO-Probes 关注 subject-verb-object 尤其 verb understanding，发现模型难以区分细粒度动作差异；Winoground 使用 1,600 text-image pairs，caption words 相同但顺序和关系不同，测试 visio-linguistic compositionality，很多模型甚至 below chance。Intrinsic evaluation 则直接看 representation quality，例如 attention patterns 是否接近人类、语义相似判断或 brain responses 是否更 aligned。

考试问法： 如果考 VLM evaluation，答案要分层：task-specific/extrinsic evaluation 看下游表现；challenge datasets 如 FOIL、SVO-Probes、Winoground 检查具体 grounding 和 compositional skills；intrinsic evaluation 检查表示是否与人类注意、语义相似或脑反应对齐。高分点是指出高 benchmark score 不等于真正 visual grounding。

9. 总收束：VLM 是通向 grounded NLP 的桥，不是终点

Summary

- Current NLP models have been trained using large-scale datasets only. This limits the variety of signal they can access to for learning.
- It has been shown that language understanding is an activity that leverages multiple parts of the brains that are involved to complete other tasks as well. The brain is **inherently multimodal**!
- How do we design Perceptually grounded NLP systems? NLP systems can live in different **world scopes** — different data distributions that an agent can leverage to learn language
- Vision+Language models are a promising avenue to bridge the gap between fully multimodal systems and text-only models

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人话直觉

这节课最后的结论不是“有图就万事大吉”。更准确地说，VLM 给 text-only NLP 架了一座桥：桥的一头是大规模文本模型，另一头是感知世界。但桥还没有通向完整的人类经验——行动、触觉、社会协作、长期记忆和真实后果都还没完全进来。

精确定义 / 机制： Lecture 25 的 summary 把主线压缩成四点。第一，当前 NLP 模型主要从 large-scale text datasets 学习，能接触的 learning signal 种类有限。第二，人类语言理解调动大脑中与感知和行动相关的多个系统，brain is inherently multimodal。第三，perceptually grounded NLP 可以被理解为把模型放进不同 world scopes：不同数据分布让 agent 学到不同语言知识。第四，Vision+Language Models 是连接 fully multimodal systems 和 text-only models 的 promising avenue。也就是说，VLM 的价值在于提供视觉 grounding 和更丰富的表示，但它仍可能受数据噪声、spurious correlations、语言 prior、缺乏行动反馈和评测不充分影响。

考试问法：如果考综合题“why VLMs matter and what are their limitations”，用这个结构答：text-only limitation → symbol grounding/reporting bias → multimodal brain and world scopes → VLM tasks/architectures/training → evaluation shows remaining gaps。这样比只背任务列表更像一份完整论述答案。

考试速记版

- Symbol grounding problem: 符号如果只由其他符号解释，意义会在 token 系统里空转，缺少 real-world referents。
- Chinese Room / dictionary-go-round 的共同点：正确符号操作或字典循环不等于真正理解。
- Perceptually grounded NLP: 让语言系统通过 perception、categorisation、interaction 把词接到对象、属性、动作和事件。
- World scope 描述 learner 可访问的数据/环境；scope 越高，能学到的知识和 reporting bias 都会改变。
- Reporting bias: 文本常省略显而易见的感知事实，图像/视频能提供文本不常说的视觉信号。

- VLM 基础任务: image captioning、VQA、visual grounding/referring expression; 扩展包括 visual storytelling 和 visual dialogue。
- CNN 提取 feature maps / pooled features; Vision Transformer 把 image patches 当作 visual tokens。
- 早期 captioning: CNN 初始化 RNN hidden state, 视觉只在开头 conditioning, 后续可能被语言 prior 带偏。
- 三种 Transformer-based fusion: cross-encoding、cross-attention、visual prefix。
- V+L losses: MIM 练视觉重构, LM 练语言, ITM 练图文匹配, MMM 练跨模态补全。
- Extrinsic evaluation 看任务表现; intrinsic evaluation 看表示质量是否接近人类语义、注意或脑反应。
- FOIL 测 noun mismatch, SVO-Probes 测 verb/action understanding, Winoground 测 compositional visual grounding。
- VLM 是从 text-only 到 grounded NLP 的重要桥梁, 但不是完整 embodied/social grounding。

一段稳妥考场回答

Vision+Language Models are motivated by the limits of text-only language learning. If symbols are learned only through other symbols, as illustrated by the Chinese Room and dictionary-go-round arguments, their meaning may not be grounded in real-world referents. Perceptually grounded NLP therefore aims to connect language to perception, categorisation and interaction. The Experience Grounds Language view describes different world scopes: a learner with only text, with images, with action or with social interaction has access to different data distributions and different knowledge, and multimodal data can reduce reporting bias in text. In practice, VLMs combine visual and linguistic inputs for tasks such as image captioning, visual question answering, visual grounding, referring expression understanding, visual storytelling and visual dialogue. Visual information can be represented with CNN features or Vision Transformer patch tokens. Transformer-based VLMs fuse modalities through cross-encoding, cross-attention or visual prefix mechanisms, usually initialised from language and vision experts and trained with losses such as language modelling, masked image modelling, image-text matching and masked multimodal modelling. Evaluation should include not only downstream task scores but also challenge datasets such as FOIL, SVO-Probes and Winoground, plus intrinsic measures of representation quality. Overall, VLMs are a promising bridge between text-only NLP and fully grounded multimodal systems, but high benchmark performance does not by itself prove genuine compositional visual grounding.